The Pauli Lightcone Frontier as a Space-Time Observable for Quantum Noise Characterization

Paolo D'Alberto

Advanced Micro Devices, Inc.
paolo.dalberto@amd.com

Abstract

We simulate the kicked Heisenberg Ising model on two lattices and study the non-identity Pauli weight $n(v,t)$, the probability that the evolved operator has support at site v at cycle t, as a space-time observable. Under noise amplification at levels γ, the $n(v,t)$ directly reveals how noise deforms the operator lightcone across the lattices. We introduce the *wavemap*: a spatial portrait of noise effects assigning each site a per-noise-level arrival delay $l_\gamma(v)$ and cross-entropy loss $\mathcal{L}_\gamma(v)$. The wavemap provides direct observable signatures of the noise eigenvalue structure: pure amplitude damping produces a uniform spatial delay with no weight injection outside the lightcone, while rotation-type noise injects weight at unvisited sites, making the frontier appear locally faster. Applying the multi-product formula (MPF) with time-adaptive coefficients $\alpha(t)$ over the frontier recovers up to 55% of the information loss relative to the best noisy sample. The noise amplification parameter γ maps directly to the gate-folding factor in zero-noise extrapolation, making the wavemap methodology immediately applicable to real quantum processors.

Introduction and Setup

We simulate the kicked isotropic Heisenberg model (Begušić and Chan 2025):

$$H = J\sum_{\langle ij\rangle}(X_iX_j + Y_iY_j + Z_iZ_j), \quad U_{\text{kick}} = \prod_v R_x(2\phi) \tag{1}$$

with $\phi = 1.414$, on two topologies: a heavy-hex 3×3 lattice (68 sites, 76 bonds) with per-bond noise PTM representative of realistic heavy-hex device noise (uniform $J = 1$ couplings), and a rectangular 7×7 lattice (49 sites, 84 bonds) with synthetic uniform depolarizing noise. We use a GPU-accelerated belief propagation tensor network simulator (D'Alberto 2026) with Trotter step $\varepsilon = 0.10$, bond dimension χ, and truncation cutoff $\delta = 10^{-3}$ ($\chi = 200$ heavy-hex, $\chi = 50$ rect 7×7). The simulator is a GPU-first implementation of the belief propagation tensor network methodology; the Julia reference implementation used in (Robertson et al. 2026) is based on ITensor (Fishman, White, and Stoudenmire 2022) and TensorNetworkQuantumSimulator (Tindall and Fishman 2023; Rudolph and Tindall 2025).

Noise enters via Pauli transfer matrices (PTM) (Angrisani et al. 2026): each bond PTM composes the ideal gate with $M_\gamma^{(b)} = \gamma M_{\text{hw}}^{(b)} + (1-\gamma)I$, amplifying hardware noise by factor $\gamma = 1, 2, 3$. The standard observable, the autocorrelation $C(t) = \langle Z_c(t)Z_c(0)\rangle$, compresses the entire space-time evolution into one scalar per cycle, losing spatial information and dominated by the lightcone interior where χ is large and the approximation is least faithful.

Related approaches exploit the lightcone causal structure for error mitigation: lightcone shading (Eddins, Tran, and Rall 2024) tightens probabilistic error cancellation bounds by restricting to the observable's causal cone; non-zero noise extrapolation (Thompson et al. 2025) adds noise to ease tensor network simulation then extrapolates back, the reverse of our amplification strategy; and Pauli propagation (Eddins et al. 2026) cancels noise classically before hardware measurement. Recent work (Chang and Ippoliti 2026) proposes exploiting the coupling between depolarizing noise and operator weight to extract a global weight spectrum from noisy circuits at variable noise strength. We extend this to a site-resolved observable: rather than integrating over all sites, we track $n(v,t)$ per site per cycle, recovering the full spatial structure of how noise deforms the lightcone. The non-identity Pauli weight $n(v,t) = \sum_{P:\, P_v \neq I} c_P(t)^2$ is extracted from the same simulation at no additional cost. By normalization $\sum_P c_P^2 = 1$, it is a probability (Nahum, Vijay, and Haah 2018) and retains the full spatial field at every cycle. At the lightcone frontier where $\chi \approx 1$ and the simulation is exact, $n(v,t)$ is the most reliable observable the simulation produces, provided the interior has sufficient χ to propagate the signal forward faithfully. Eigenvalue analysis of the composed PTM confirms both noise models are in the pure amplitude damping regime (off-diagonal fraction $\approx 36\%$, constant across γ, $\Delta < 0.002$): noise delays and damps the Hamiltonian-determined wave without redirecting it, making the wavemap a model-free noise diagnostic. This is consistent with the analytical result of (Yu, He, and Wu 2026) that a noise event contributes only when it overlaps the Heisenberg evolution of the measured Pauli operator; our wavemap makes this overlap structure spatially explicit and site-resolved. Fitting time-adaptive MPF coeffi-

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cients $\alpha(t)$ over the frontier recovers up to 55% of information loss on heavy-hex and 65% on rect 7×7; since γ maps to the ZNE gate-folding factor (Giurgica-Tiron et al. 2020), the method applies directly to real quantum processors.

The Pauli Lightcone Wavemap

This section introduces the wavemap observables following the progression:

$$C(t) \xrightarrow{\text{spatialize}} n(v,t) \xrightarrow{\text{amplify}} n_\gamma(v,t) \xrightarrow{\text{minimize}} \min_\alpha \mathcal{L} \tag{2}$$

Autocorrelation as observable

The standard observable in tensor network simulation of quantum lattice models is the autocorrelation $C(t) = \langle Z_c(t) Z_c(0) \rangle$, computed as an inner product between the evolved operator and the initial state at the center site c. This compresses the entire space-time evolution into one number per cycle. Sites strongly affected by noise and sites barely affected contribute equally; furthermore, the computation is dominated by the interior of the lightcone, where bond dimension χ is large and the tensor network truncation is least faithful.

The Pauli weight $n(v,t)$ as a spatial observable

Any operator $O_c(t)$ initialized at the center and expanded in the n-qubit Pauli basis $\{I,X,Y,Z\}^{\otimes n}$ as $O_c(t) = \sum_P c_P(t)\, P$ satisfies $\sum_P c_P(t)^2 = 1$. The non-identity Pauli weight at site v and cycle t is:

$$n(v,t) = \sum_{P:\, P_v \neq I} c_P(t)^2 \tag{3}$$

The same simulation that computes $C(t)$ also computes $n(v,t)$ at no additional cost, retaining the full spatial distribution of operator support across the lattice at every cycle.

The $n(v,t)$ as a probability

By normalization, $n(v,t) \in [0,1]$ is the probability that a Pauli string sampled from the operator distribution acts non-trivially at site v (Nahum, Vijay, and Haah 2018; Tan and Brouwer 2025). This is not an approximation: it follows directly from $\sum_P c_P^2 = 1$. The field $n(v,t)$ is therefore a probability distribution over the lattice, evolving in time as the operator spreads outward from the initialization site.

Lightcone: velocity and information

The *frontier* at cycle t is the set of sites first crossing a significance threshold $\varepsilon_{\text{th}} = 10^{-4}$:

$$\mathcal{F}(t) = \{v : n(v,t) > \varepsilon_{\text{th}},\ n(v,t-1) \leq \varepsilon_{\text{th}}\} \tag{4}$$

Its maximum breadth-first search (BFS) distance grows linearly at velocity v_{hop} (hops/cycle), establishing the Lieb-Robinson cone. The wavemap observables are computed at the leading edge of the lightcone where they are least sensitive to bond dimension truncation. The threshold ε_{th} plays the role of a physical tolerance — analogous to the truncation cutoff δ in the simulation — defining the precision with which we observe the wavefront's presence and amplitude ratio. The choice of χ reflects three practical constraints: memory, wall-clock time, and accuracy; its effect on $C(t)$ convergence and collapse was studied separately in (D'Alberto 2026). A full characterization of the interplay between χ, ε_{th}, and wavemap accuracy is left for future work. Crucially, the simulation captures non-uniform frontiers naturally: irregular topology, per-bond coupling heterogeneity, and asymmetric noise all shape $\mathcal{F}(t)$ without any model assumption, as the heavy-hex results demonstrate (near-degenerate $v_{\gamma_2} \approx v_{\gamma_3}$, non-monotone delay, Table 2).

Noiseless and noisy waves

We simulate four variants: noiseless (NL) and three noise amplification levels $\gamma = 1, 2, 3$ (chosen from the ZNE literature (Giurgica-Tiron et al. 2020)), where γ rescales the noise channel $M_\gamma = \gamma M_{\text{hw}} + (1-\gamma) I$. Each variant produces its own wavefront advancing at velocity $v_\gamma \leq v_{\text{nl}}$: noise suppresses amplitudes below threshold, delaying the frontier crossing. For each site v and noise level γ we define the *arrival delay* $l_\gamma(v) = t_\gamma(v) - t_{\text{nl}}(v) \geq 0$ and the *amplitude ratio* $r_\gamma(v) = n_\gamma(v, t_\gamma(v)) / n_{\text{nl}}(v, t_\gamma(v)) \leq 1$, both guaranteed by the Lieb-Robinson bound for pure amplitude damping. Together they form the *wavemap*: a spatial portrait of how noise deforms the lightcone, site by site and cycle by cycle (see Appendix, Figure 1).

Cross-entropy as the information cost

Since $n(v,t)$ is a probability, the natural measure of deviation between the noisy and noiseless is by loss of information. As Jaynes (Jaynes 1957a,b) established, when quantities are probabilities, information-theoretic objectives are the correct and unifying choice: MSE penalizes amplitude error uniformly, while cross-entropy penalizes information loss proportionally to its magnitude. We define the per-site cross-entropy loss summed over a delay-adaptive window $[t_\gamma(v) - \Delta,\, t_\gamma(v) + \Delta]$, $\Delta = \max(1, l_\gamma(v))$:

$$\mathcal{L}_\gamma(v) = \sum_{t \in \text{window}} -\ln \frac{n_\gamma(v,t)}{n_{\text{nl}}(v,t)} \tag{5}$$

Zero when the distributions coincide, positive when noise suppresses probability, with window width set by the delay itself so no free parameter is introduced. This is the accumulated information loss at site v: the wavemap observable that quantifies, site by site, how much the noisy evolution has forgotten relative to the noiseless reference.

The next section shows how these observables enable MPF error recovery.

Results

Following the progression of Eq. 2, we present results in three stages: the autocorrelation $C(t)$, the frontier velocity v_{hop} from $n(v,t)$, and the MPF entropy recovery.

Stage 1: Autocorrelation $C(t)$

Existing MPF methods operate at two levels of the same scalar observable. The first fits $C(t)$ directly: $\sum_i c_i C_{\gamma_i}(t) \approx$

$C_{\mathrm{nl}}(t)$, one equation per cycle. The second solves the bilinear system $A\alpha = B$ (Robertson et al. 2025, 2026), where $A_{ij} = \langle\psi_{\gamma_i}|\psi_{\gamma_j}\rangle$ are noisy-noisy inner products and $B_i = \langle\psi_{\gamma_i}|\psi_{\mathrm{nl}}\rangle$ are noisy-noiseless inner products: a richer fit, but still a global scalar per cycle, still dominated by the interior of the lightcone where χ is large and the approximation is least faithful.

Table 1 shows $C(t)$ at each cycle for NL and $\gamma = 1, 2, 3$ on both topologies. On heavy-hex, $C(t)$ oscillates and the noise levels are nearly indistinguishable: the scalar observable is almost blind to noise. On rect 7×7, $C(t)$ decays monotonically and the γ levels separate, but all spatial structure is lost. In both cases, $C(t)$ and $A\alpha = B$ compress the entire space-time evolution into one number per cycle.

Stage 2: Frontier velocity from $n(v, t)$

The non-identity Pauli weight $n(v, t)$ resolves the spatial structure invisible to $C(t)$. Table 2 shows the frontier velocity v_{hop} (hops/cycle), the slope of the maximum BFS frontier distance versus cycle.

The frontier velocity v_{hop} is not a property of the noise alone. It is a joint function of three factors: the graph topology (minimum hop distance between connected sites, coordination number, girth), the Hamiltonian coupling J_{ij} (which determines how much Pauli weight transfers per gate application), and the noise representation at gate level (how γ rescales the effective amplitude per bond crossing). The per-site delay statistics (n_{reached}, μ_l, σ_l) in Table 2 provide the statistical fingerprint of all three combined: sites that the noisy wave fails to reach entirely ($n_{\mathrm{reached}} < n_{\mathrm{nl}}$), the mean delay for those it does reach (μ_l), and the spread (σ_l) reflecting path-dependent variation in coupling and noise accumulation. The wavemap therefore characterizes the interplay of geometry, dynamics, and noise in a single measurement.

Sites outside the noiseless lightcone have $\chi = 1$ identically: the operator has not yet reached them and they carry no correlations. Simulation outside the cone is unnecessary: the arrival time $t_{\mathrm{nl}}(v) = d(c, v)/v_{\mathrm{hop}}$ provides a per-site entry schedule, enabling lightcone-aware memory allocation that eliminates computation before each site activates. Crucially, the wavemap turns this boundary into a diagnostic: any $n_\gamma(v, t) > \varepsilon_{\mathrm{th}}$ observed *outside* the noiseless cone is not silence but a signal: spontaneous activation that the frontier flags as anticausal. For pure amplitude damping this never occurs ($l_\gamma(v) \geq 0$ always); its presence is a fingerprint of rotation-type noise injecting Pauli weight independently of the Hamiltonian causal structure.

Stage 3: MPF entropy recovery

The wavemap cross-entropy loss $\mathcal{L}_\gamma(v)$ is both a diagnostic and an objective: minimizing it over MPF coefficients recovers the noiseless Pauli weight field from noisy samples.

All methods reconstruct $n_{\mathrm{MPF}}(v, t) = \sum_i c_i\, n_{\gamma_i}(v, t)$ and are evaluated by the global mean cross-entropy loss $\mu_{\mathcal{L}}$ (Eq. 5), lower is better.

A geometric baseline fits c by minimizing squared error on the frontier distance series $\min_c \sum_t (F_{\mathrm{MPF}}(t) - F_{\mathrm{nl}}(t))^2$, subject to $F_{\mathrm{MPF}}(t) \leq F_{\mathrm{nl}}(t)$. This objective sees only scalar distances, never amplitudes. Evaluating its n_{MPF} on $\mu_{\mathcal{L}}$ yields 1.058/3.911 with no improvement over the best noisy sample, confirming that geometric and information-theoretic objectives are not aligned.

MPF entropy fits c by directly minimizing $\mu_{\mathcal{L}}$:

$$\min_{c_i} \sum_{v,t} -\ln \frac{n_{\mathrm{MPF}}}{n_{\mathrm{nl}}}, \quad \sum_i c_i = 1, \quad n_{\mathrm{MPF}} \leq n_{\mathrm{nl}} \tag{6}$$

Table 3 shows $\mu_{\mathcal{L}}$ for the noisy samples and MPF entropy on both topologies at 10 cycles; the geometric baseline is excluded as it optimizes a different objective.

Time-adaptive coefficients $\alpha(t)$

The global MPF entropy fit uses fixed c across all space and time. We refine this by fitting one coefficient vector $\alpha(t) \in \mathbb{R}^3$ per cycle over the union frontier $\mathcal{F}(t) = \mathcal{F}_{\mathrm{nl}}(t) \cup \bigcup_i \mathcal{F}_{\gamma_i}(t)$, a thin spatial shell where $\chi \approx 1$ in all simulations and every sample is exact. The approximation is in the application: coefficients fitted on the frontier are extrapolated to all sites at that cycle.

The dual, $\alpha(d)$, fixes a BFS shell and fits over a time window, a Δ in time rather than space. Table 4 shows $L(t) = \sum_v -\ln(n_{\mathrm{MPF}}/n_{\mathrm{nl}})$, the global cross-entropy loss summed over active sites at cycle t, for $\gamma = 1$, $\alpha(t)$, and $\alpha(d)$: heavy-hex at 20 cycles (to expose the frontier collapse at cycles 12–13), rect 7×7 at 10 cycles. Table 5 shows $L(d)$ per BFS shell at 10 cycles for both topologies.

Connection to hardware: ZNE

The noise amplification parameter γ maps directly to the gate-folding factor in zero-noise extrapolation (ZNE) (Giurgica-Tiron et al. 2020): running the circuit at fold factors $\gamma = 1, 2, 3$ multiplies the effective noise per gate by γ, exactly the M_γ model above. The Pauli weight $n(v, t)$ is accessible per qubit per cycle via local Pauli measurements. On real quantum hardware there is no bond dimension: the full quantum state is available exactly, and $n(v, t)$ is measured directly without truncation artifact. The frontier is then the leading edge of the measured operator support, defined by the same threshold $\varepsilon_{\mathrm{th}}$ but exact at every site. The wavemap observables and $\alpha(t)$ fitting then follow from the same computation applied to experimental data, no simulation required. The present work provides the ground truth and the methodology; hardware validation is a direct next step.

Conclusion

A brief list of consideration follows: First, eigenvalue analysis of the composed gate-plus-noise PTM confirms pure amplitude damping: the spatial propagation pattern is entirely a property of the Hamiltonian, validating the wavemap as a model-free noise diagnostic. Second, the Lieb-Robinson constraint $n_{\mathrm{MPF}} \leq n_{\mathrm{nl}}$ is a hard causal bound, not a regularizer; enforcing it pointwise is an open problem whose solution would eliminate the floored entries in Table 4. Third, fitting time-adaptive coefficients $\alpha(t)$ over the union frontier recovers 55% of information loss on heavy-hex and 65% on rect 7×7, outperforming both fixed global coefficients and

t	Heavy-hex 3 × 3 NL	γ_1	γ_2	γ_3	Rect 7 × 7 NL	γ_1	γ_2	γ_3
1	−.924	−.924	−.925	−.925	−.914	−.916	−.918	−.920
2	+.832	+.836	+.839	+.843	+.827	+.843	+.857	+.870
3	−.763	−.773	−.783	−.792	−.748	−.791	−.826	−.854
4	+.789	+.803	+.816	+.828	+.612	+.696	+.759	+.808
5	−.828	−.848	−.864	−.878	−.500	−.623	−.711	−.774
6	+.851	+.876	+.895	+.909	+.414	+.574	+.685	+.761
7	−.838	−.870	−.892	−.907	−.341	−.534	−.665	−.751
8	+.817	+.857	+.882	+.897	+.287	+.508	+.654	+.744
9	−.804	−.853	−.881	−.896	−.248	−.495	−.650	−.742
10	+.793	+.853	+.883	+.898	+.219	+.490	+.649	+.740

Table 1: $C(t)$ for NL and $\gamma = 1, 2, 3, 10$ cycles ($\varepsilon_{\text{th}} = 10^{-4}$). Heavy-hex: noise levels nearly indistinguishable. Rect 7 × 7: monotone decay, spatial information lost. χ grows $4 \to 163$ on heavy-hex; saturates at $\chi = 50$ on rect 7 × 7 (truncation error present at interior sites; wavemap observables computed at frontier where $\chi \approx 1$ and simulation is exact.)

			$\gamma = 1$			$\gamma = 2$			$\gamma = 3$		
Topology	v_{nl}	n_{nl}	n	μ_l	σ_l	n	μ_l	σ_l	n	μ_l	σ_l
HH 3 × 3	0.376	21	8	1.13	1.97	10	2.10	2.81	8	0.75	1.30
Rect 7 × 7	0.448	49	49	0.98	1.15	37	1.35	1.36	25	1.24	1.77

Table 2: Frontier velocity v_{nl} and delay statistics (n, μ_l, σ_l) per γ: sites reached, mean and std of arrival delay $l_\gamma(v)$ in cycles.

Method	μ hh dep	μ rect 7×7
$\gamma = 1$	1.058	3.680
$\gamma = 2$	1.538	7.432
$\gamma = 3$	2.277	8.003
MPF entropy	0.596	2.783

Table 3: Mean per-site $\mu_{\mathcal{L}}$ (lower is better), 10 cycles. MPF entropy reduces loss 44% on heavy-hex and 24% on rect 7 × 7.

space-adaptive $\alpha(d)$. Fourth, fast GPU-accelerated simulation enables systematic exploration of the causal relations between noise character, Hamiltonian coupling, and wavefront structure across a library of noise models and topologies. This opens the path to classifiers trained on simulation ground truth that diagnose noise type and severity directly from measured $n(v, t)$ on real quantum processors, closing the loop between classical simulation and hardware characterization.

Wavemap Visualization and Entropy Recovery

This appendix provides the spatial visualization of the wavemap and the MPF entropy recovery that complements the numerical results in Tables 1–5.

Figure 1 shows two key aspects simultaneously. **Top panel**: the arrival delay $l_\gamma(v)$ and amplitude ratio $r_\gamma(v)$ for $\gamma = 1, 2, 3$ on the heavy-hex 3×3 lattice. White sites are outside the noiseless lightcone; black sites were reached by the noiseless wave but not by the noisy wave within 20 cycles.

t	Heavy-hex γ_1	$\alpha(t)$	$\alpha(d)$	Rect 7 × 7 γ_1	$\alpha(t)$	$\alpha(d)$
1	0.11	0.00	0.00	0.85	0.01	0.01
2	0.23	0.02	0.00	3.17	0.25	0.66
3	0.91	0.16	0.00	8.83	1.56	2.00
4	1.33	0.29	0.00	20.00	5.42	18.88
5	1.61	0.42	0.00	33.51	12.16	39.20
6	3.58	1.22	2.16	43.47	18.45	53.56
7	5.30	2.18	4.90	44.96	19.46	55.85
8	6.76	3.43	6.86	46.06	20.64	57.51
9	8.11	4.89	8.03	47.41	21.69	59.26
10	8.58	5.44	8.47	49.06	23.06	61.16
11	14.90	14.08	18.33	–	–	–
12	22.12	26.26	29.75	–	–	–
13	23.51	*41.83*	31.81	–	–	–
14	24.69	28.76	33.47	–	–	–
15	27.60	32.36	37.88	–	–	–
16	28.65	**7.35**	39.38	–	–	–
17	38.48	12.77	53.10	–	–	–
18	39.36	14.34	54.15	–	–	–
19	42.89	16.72	58.87	–	–	–
20	50.29	21.62	68.60	–	–	–
μ	1.058	**0.481**	1.050	3.680	**1.276**	5.808

Table 4: $L(t) = \sum_v -\ln(n_{\text{MPF}}/n_{\text{nl}})$: global cross-entropy loss per cycle. Hh at 20 cycles, r7 at 10 (– = not run). *Italics*: $\alpha(t)$ ill-conditioned (frontier collapse, cycles 12–15). **Bold**: recovery after frontier repopulates. μ row: mean $\mu_{\mathcal{L}}$. Values floored at zero.[1]

Colored sites show the delay in cycles (red = large delay) or amplitude ratio (green = ratio near 1, red = strongly attenuated). On the heavy-hex lattice with hardware-characterized noise, the delay is non-monotone in γ — reflecting the rotation content of the PTM which can inject weight at previously unvisited sites, locally accelerating the apparent frontier. On the rect 7 × 7 lattice with synthetic uniform depolarizing noise, the delay is monotone in γ as expected for pure amplitude damping. **Bottom panel**: the MPF entropy recovery. **Row 1** (red): per-site cross-entropy $\mathcal{L}(v)$ for noisy $\gamma = 1, 2, 3$ — each column shows a spatial snapshot of

	Heavy-hex			Rect 7 × 7		
d	γ_1	$\alpha(t)$	$\alpha(d)$	γ_1	$\alpha(t)$	$\alpha(d)$
0	0.00	0.00	0.00	0.00	0.00	0.00
1	0.03	0.00	0.00	0.07	0.00	0.00
2	0.07	0.01	0.00	0.25	0.02	0.01
3	0.21	0.05	0.00	0.62	0.13	0.10
4	0.51	0.24	0.92	1.11	0.40	1.79
5	0.88	0.62	1.50	1.70	0.84	2.53
6	–	–	–	2.33	1.38	3.27

Table 5: Mean cross-entropy loss $\mathcal{L}(d)$ per BFS shell. $\alpha(t)$ reduces loss at every shell; $\alpha(d)$ degrades at outer shells (too few sites per shell to constrain the fit).

information loss at the frontier. **Row 2** (green): the same quantity after MPF recovery with $\alpha(t)$ coefficients — the information loss is visibly reduced across all frontier sites. **Row 3**: integrated loss $L(t) = \sum_v \mathcal{L}(v)$ per cycle (noisy in red, MPF-recovered in green). The $\alpha(t)$ optimizer is ill-conditioned at cycles 12–15 (frontier collapse onto a thin shell) but recovers at cycle 16 when the frontier expands again. **Row 4**: mean $\mathcal{L}(v)$ per BFS distance shell — $\alpha(t)$ consistently reduces the cross-entropy at every shell, confirming that the recovery is spatially uniform, not concentrated at particular sites.

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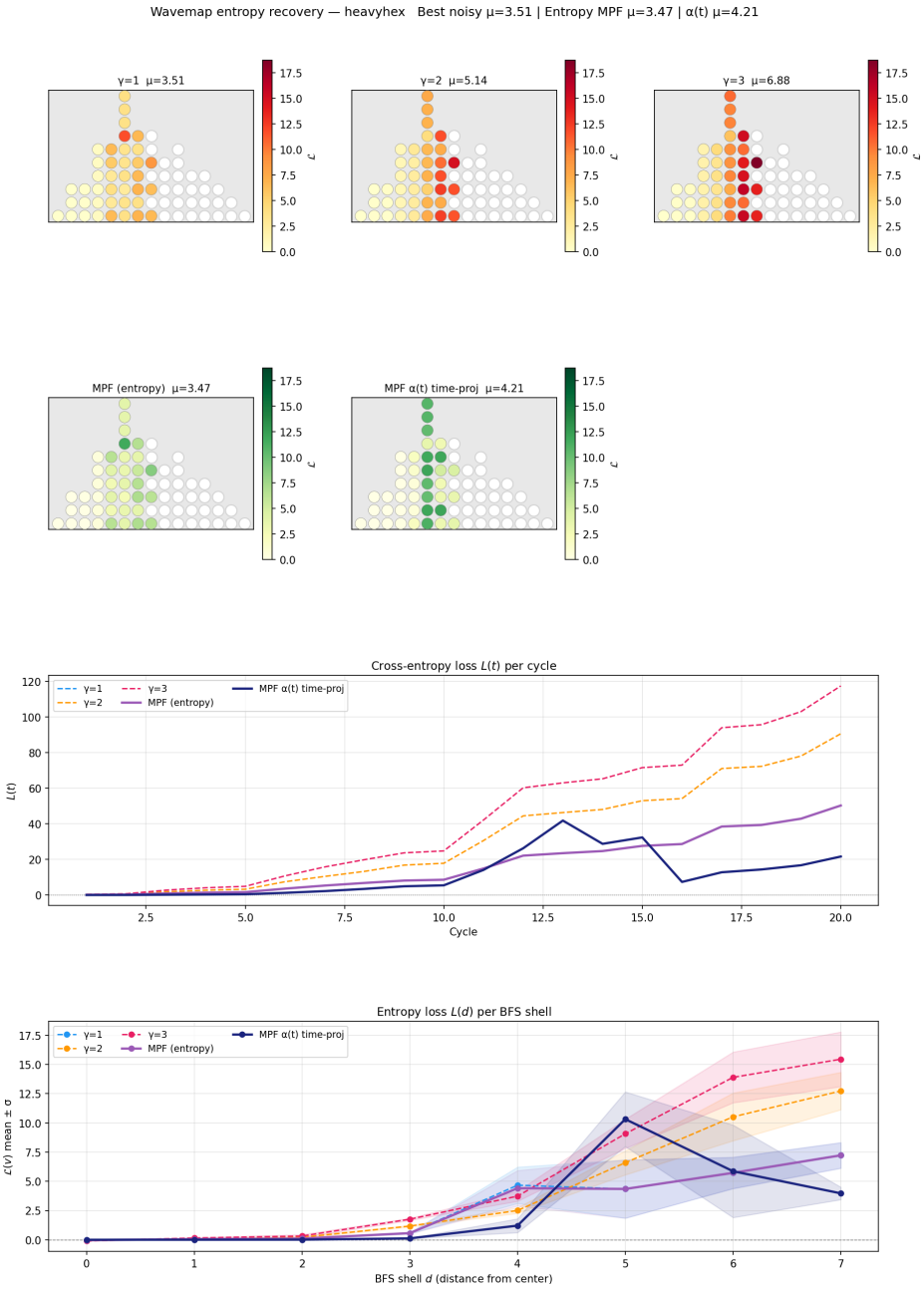


Figure 1: Wavemap visualization and entropy recovery, heavy-hex 3×3, 20 cycles ($\chi = 200$, $\delta = 10^{-3}$, $\varepsilon_{\text{th}} = 10^{-4}$). **Top**: arrival delay $l_\gamma(v)$ (left) and amplitude ratio $r_\gamma(v)$ (right) for $\gamma = 1, 2, 3$. White: outside lightcone. Black: reached by noiseless wave only. Non-monotone delay on heavy-hex reflects hardware PTM rotation content. **Bottom, Row 1** (red): per-site $\mathcal{L}(v)$ for noisy runs. **Row 2** (green): $\mathcal{L}(v)$ after $\alpha(t)$ MPF recovery — visibly reduced. **Row 3**: $L(t) = \sum_v \mathcal{L}(v)$ per cycle; ill-conditioned at cycles 12–15 (frontier collapse), recovers at cycle 16. **Row 4**: mean $\mathcal{L}(v)$ per BFS shell; $\alpha(t)$ reduces loss at every shell.